

Mukul Ved

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Summary

Data professional with around 3 years of experience in **data engineering, analytics, and applied machine learning** across consulting, IT, and non-profit domains. Proficient in **Python, SQL, R, PySpark, Airflow, dbt, and cloud platforms (AWS, Azure, BigQuery)** to build ETL pipelines and compliance-ready data solutions. Delivered 25+ dashboards in **Tableau, Power BI, and Looker**, empowering executives and teams with insights that improved financial decisions, streamlined operations, and advanced education outcomes.

Professional Experience

Data Analyst • Chicago Education Advocacy Cooperative - Chicago, IL APR 2025 - Present

- Engineered predictive models in **Python and SQL**, integrating **10,000+ internal survey responses** with **public socioeconomic datasets** to flag **high-risk student disengagement cases** and enable targeted interventions.
- Standardized heterogeneous data formats (**scaled, binary, open-text**) with **Pandas and NumPy**, collaborating with academic researchers to ensure **machine learning readiness** for **5 longitudinal study cohorts**.
- Applied **NLP and sentiment analysis** techniques to **3,500+ qualitative responses**, surfacing **emotional and behavioral signals** that correlated with **dropout risks** and informing **curriculum redesign** for **12 partner schools**.

Data Analyst Intern • Excelrate.org - USA AUG 2023 - OCT 2023

- Directed a **cross-functional team of 16 members**, coordinating **advertising performance analysis** with **Excel and SQL**, and uncovering spend inefficiencies that lowered ad costs by **\$42K annually** while expanding reach to **5 new regions**.
- Designed and automated **Tableau dashboards** to monitor **campaign effectiveness**, shortening reporting cycles by **14 hours per month** and enabling executives to review **real-time insights** on survey quality and transparency.
- Collaborated with **HR and finance stakeholders** to track **payroll-linked advertising expenses**, using **data modeling in Excel** to reallocate **\$18K in budget** toward **high-performing campaigns** and improve **financial planning accuracy**.

Data Analyst • Dell Technologies - India JUN 2022 - MAY 2023

- Constructed **batch and streaming data pipelines** with **PySpark, Apache Kafka, and Airflow**, processing **25M+ device telemetry and support ticket records monthly** to enable **near real-time operational analytics**.
- Integrated datasets from **AWS S3 and on-prem SQL Server** into a **unified data lake**, consolidating **global IT assets across 190+ regions** and reducing **complex query latency from 22 seconds to under 7 seconds**.
- Optimized **revenue and warranty analytics workflows** with **SQL, Pandas, and dbt**, surfacing trends in **service contract renewals and hardware replacement cycles**, influencing **\$68M+ in support revenue**.
- Developed **compliance-ready storage solutions** in **PostgreSQL and MongoDB**, collaborating with **governance teams** to manage retention of **12M+ customer and device records** aligned with **GDPR and HIPAA standards**.
- Designed and delivered **15 Tableau and Power BI dashboards** visualizing **supply chain, warranty claims, and capacity forecasts**, enabling **global operations managers** to reallocate inventory across top product markets and improve **asset utilization**.

Data Analyst | KPMG - India MAY 2021 - MAY 2022

- Engineered **ETL pipelines** with **PySpark and Apache Airflow**, integrating **18M+ financial and operational records monthly** into a **centralized data warehouse**, reducing **reporting latency from hours to under 5 seconds** for audit teams.
- Synthesized **client feedback and survey datasets** using **Python, Pandas, and SQL**, applying **A/B testing and ANOVA** to surface **insights** that improved **customer satisfaction scoring by 1.8 points** across consulting engagements.
- Designed and delivered **12 interactive dashboards** in **Power BI and Tableau**, tracking **regional revenue, workforce utilization, and compliance KPIs**, enabling executives to shorten **decision cycles by 2.4 days**.
- Modeled **structured and unstructured datasets** in **PostgreSQL and MongoDB**, enabling **predictive analysis on project risks** and reducing **escalation cases by 1,200 annually** across enterprise accounts.
- Orchestrated **AWS S3-based data lakes** with **dbt transformations**, partnering with **compliance and IT teams** to streamline **audit deliverables** and improve **reporting efficiency by 15 hours per cycle**.

Technologies & Skills

- Programming & Scripting:** Python, R, SQL, JavaScript (ES6+)
- Business Intelligence & Visualization:** Tableau, Power BI, Looker, Matplotlib, Seaborn, Excel
- Data Engineering & Big Data:** PySpark, Hadoop (HDFS, HBase), Apache Kafka, Apache Airflow, dbt, ETL
- Machine Learning & Analytics:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Large Language Models (LLMs), A/B Testing, Hypothesis Testing, Time Series Analysis, ANOVA
- DevOps & Tools:** Git, Docker
- Cloud Platforms:** AWS (S3, SageMaker), Azure, Google Cloud (BigQuery)
- Databases:** PostgreSQL, MySQL, MongoDB

Education

- ILLINOIS INSTITUTE OF TECHNOLOGY**, Master of Computer Science AUG 2023 – MAY 2025
- Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Data Analysis, Cloud Computing, Big Data Analytics, Computer Networks, Advance Database Organization, Digital Healthcare Inform & AI, Data Mining, Natural Language Processing .
- GAUTAM BUDDHA UNIVERSITY**, Bachelor's in Computer Science and Engineering | INDIA AUG 2019 – JUN 2023

Certifications

- [Google Data Analytics Professional Certificate](#)
- [Codecademy - Data Scientist Professional](#)
- [Codecademy - Business Intelligence Data Analyst](#)
- [IBM Applied AI Professional](#)